"Can machine learning models predict whether a given season is a batter's peak year, based on prior performance data?"

CONTENTS:

INTRODUCTION

Talent scouting and recruitment are undergoing transformations, with data science and machine learning (ML) offering a revolutionary approach to player evaluation [1]. Traditionally, player recruitment has relied heavily on subjective assessments, intuition, and personal bias, often leading to inconsistent decisions [2]. By integrating advanced analytics with quantitative player statistics, teams can adopt a more objective, data-driven approach to identifying and nurturing talent, ultimately improving squad selection and performance [3].

The rise of performance databases, such as ESPNcricinfo [4], offer easy access to detailed player statistics, aiding franchises, in leagues like the Indian Premier League (IPL), in making informed decisions while saving time and resources [5]. When combined with ML techniques, this data helps identify patterns and traits linked to long-term success.

Player recruitment is critical in professional sport, with poor choices leading to both performance and financial setbacks [6]. ML has been shown to help uncover hidden talent [7], a valuable asset in high-stakes T20 leagues like the IPL, where key overs often decide matches. Despite the increasing adoption of analytics in cricket, for example, in performance evaluation [8], [9] and team selection [3], [10], predicting the exact timing of peak performance remains understudied. The ability to identify when a player is likely to hit peak form can inform vital decisions around team selection and retention. This report addresses this gap, using supervised learning techniques on IPL batting data, to predict a batter's peak season, defined by their highest strike rate, a metric prioritised in T20 cricket for its direct correlation with match impact [11].

RESEARCH QUESTION

To explore the above, the following research question is posed: *"Can machine learning models predict whether a given season is a batter's peak year, based on prior performance data?"*

This research builds upon earlier studies that predict rising stars using Support Vector Machines (SVMs) [12] and more recent talent identification frameworks such as Khan and Ahamad's fuzzy logic-based approach (92.8% accuracy) [13]. However, the use of fuzzy logic introduces subjectivity [14]. Additionally, both studies relied on broad criteria, with the most influential factors primarily linked to fixed averages and throwing ability, respectively. This report refines the focus by objectively identifying key traits that predict success in batters, which is an important need in short-format leagues like the IPL, where squad turnover is high.

Furthermore, existing literature often uses static clustering techniques based on metrics like strike rate, assuming performance remains constant over time [3]. In contrast, the current study adopts a novel dynamic predictive framework, modelling how player performance evolves to forecast peak form, an area where ML has shown superior performance [15]. This shift to predictive analytics offers greater strategic value for scouting in professional T20 leagues.

This report hypothesises that ML can detect meaningful trends in cricket players' performance trajectories to predict their peak strike rate seasons in the IPL. By analysing multi-season batting performance, it provides evidence-based insights for player evaluation, informed recruitment strategies and performance monitoring. The methodology demonstrates how data analytics can be effectively integrated into elite sporting operations, ultimately enhancing decision-making throughout the IPL.

DATA

This analysis relies on a dataset obtained from Kaggle, a leading platform in the data science and research community, and renowned for providing high-quality, diverse datasets [16]. The dataset used [17] is a direct extraction from the Statsguru section of ESPNcricinfo [4]. It contains extensive match and player data, with this report utilising the 'Indian Premier League (IPL) Player Stats – 2016 till 2019.csv' file. This dataset contains IPL player data from 2016 to 2019, with 631 observations, 29 variables, and 268 unique players. Key metrics include the number of innings batted, total runs scored, batting average and strike rate. These variables are essential for developing a comprehensive picture of cricket player dynamics and for building predictive models.

To ensure the reliability and usability of the dataset, thorough preprocessing was conducted using R Studio (Version 2024.12.1+563). As the focus of this report is on batters, the dataset was filtered to retain only batting-related variables to eliminate any interference from irrelevant bowling metrics, which reduced the dataset to 14 variables.

Missing values marked as '-', were replaced with 'NA' for compatibility with R. Asterisks (*) in highest scores, indicating a 'not out' status, were removed to enable numerical analysis. The 'IPL' prefix in the 'Tournament.Year' variable was stripped to convert it to numeric. All variables were then formatted to appropriate data types.

While the dataset provides a robust overview of IPL batting performance across four seasons, it does have limitations, including the fact its temporal scope ends in 2019, and its exclusion of bowling and fielding metrics. Despite this, the dataset is largely complete and well-structured, making it suitable for predictive analysis of batting performance.

METHODS

RATIONALE

Given the temporal, player-specific, and performance-driven nature of the data, a modelling approach was required that could capture nonlinear interactions, handle noisy features with minimal imputation, and operate robustly with moderate sample sizes without assuming normality and homoscedasticity. Random Forests (RFs) were chosen for their ability to generate an ensemble of decision trees, which are aggregated to produce a more accurate and reliable forecast that avoids overfitting [6], [18]. This approach is valued for its computational efficiency and operational simplicity [19]. Although RFs assume feature independence, this was addressed using lagged and cumulative features that were engineered to encode temporal dependencies.

On the other hand, model stability is further enhanced with sufficient data and appropriate hyperparameter tuning [20]. To satisfy these conditions, this study employed cross-validation, feature engineering, and comprehensive preprocessing. A limitation of ensemble methods is the challenge of model interpretability [21], however, feature importance analysis was applied to increase transparency, which is critical when communicating insights to coaches who need to understand which performance indicators signal peak potential.

RF models have demonstrated predictive accuracy in various sport contexts. For instance, in esports, an RF classifier demonstrated high accuracy in predicting match outcomes and identifying the most influential performance indicators [22]. Meng [23] corroborates these findings in football, revealing RF to identify the most important features that accurately predict the outcome of football games. Furthermore, in cricket-specific research, RF outperformed decision trees and SVM, achieving an accuracy of 92.25%, thereby aiding team managers in optimising player selection and improving overall team performance [24].

ALTERNATIVE METHODS

Since the peak year was predefined, a supervised learning approach was appropriate [25]. Without this, unsupervised methods like clustering could have been employed to group seasons into categories like “peak”, “average” and “slump”. This approach has been applied in cricket, where four distinct groups of bowlers were identified using k-means cluster analysis [26]. However, K-means can converge to suboptimal solutions due to sensitivity in initial conditions [27], whereas RFs use randomness and majority voting to avoid such pitfalls.

Another supervised ML technique is SVM, which determines the optimal hyperplane for classifying and separating the data into two distinct classes [28]. SVM has demonstrated high accuracy (96.4%) in classifying cricketers as talented or not-talented [28]. However, SVM does not provide clear insights into the physiological factors contributing to these classifications, highlighting its lack of interpretability. This reinforces the choice to use RFs as they can provide feature importance, offering valuable insight for coaches. Furthermore, while SVM requires careful kernel tuning, RF performs well without extensive parameter adjustments [29]. SVMs also struggle to handle noise interference [30] and since the data in the current study was noisy, RFs were a more suitable choice.

Finally, the powerful use of neural networks were considered given their capacity to model complex, nonlinear relationships and process vast volumes of unstructured data [31]. They have shown insightful guidance when making team selection decisions in cricket [32], and graph neural networks achieved a notable 71.54% accuracy in basketball performance prediction [33]. However, those studies benefitted from substantially larger datasets (e.g., 14,758 observations [33]), compared to the moderate sample size of the current study. Due to their high data demands, greater computational complexity, and longer training times [34], neural networks were deemed less suitable in this context.

DATA CLEANING AND PREPARATION

Data cleaning ensures quality data that enhances analysis and boosts ML performance [35]. Handling missing values is a crucial aspect of data cleaning, as they can significantly impair classification accuracy by reducing the information available to ML models during training [36]. Sixty-two records with missing 'Batting.Innings' data, which affected related batting variables, were removed since these players did not bat and thus lacked relevant data. Additional missing values in 'Batting.Average' (68) and 'Batting.Strike.Rate' (8) were also excluded. As batting average cannot be reliably imputed and often indicades a player was never dismissed, removing it preserved the integrity of the target variable. Imputing these values could misrepresent player performance and introduce bias into the analysis. Outliers were reviewed and retained, as they reflected real-world performance, and RFs are inherently robust to such variation [37]. This ensured external validity as the model can learn from the full range of real-world player performance.

The target variable 'IsPeak' was operationalised by identifying each player's maximum strike rate across all seasons in the dataset, an appropriate metric for the fast-paced T20 format of the IPL due to its valuable insight into a player's scoring aggression and efficiency [9], [11]. Seasons corresponding to this peak strike rate were labelled as 1, while all other seasons were labelled as 0. Players whose peak occurred in their first recorded season were excluded due to a lack of prior performance data to inform predictive modelling.

To enhance the predictive capacity of the dataset, a custom feature engineering function was implemented in R. The goal was to add construct validity by generating player-specific temporal features that capture historical performance trends and cumulative career statistics. It was assumed that player-season records are conditionally independent and that historical performance trends are indicative of future peak seasons. Player-season records were sorted chronologically and for key batting statistics, lag features (values from the previous one and two seasons) and trend features (year-over-year change), indicating performance improvement or decline were computed. Additional cumulative career statistics up to the current season year were also derived, with missing values due to a lack of history imputed with zero. An experience metric was created to reflect the number of years a player had participated. These features were designed to capture both short-term momentum and long-term progression, yielding a final dataset of 378 observations across 44 variables.

To prevent data leakage, non-predictive and target-related columns were excluded from training. The target variable 'IsPeak' was retained separately as the response vector 'y'. The dataset was partitioned into training and test sets using an 80/20 split and a fixed random seed (42) was applied to ensure reproducibility of the split. This analysis assumes temporal stability in key performance indicators, allowing models trained on 2016-2019 data to be applicable for forecasting future player outcomes.

LOGISTIC REGRESSION

A logistic regression model was developed as a baseline due to its interpretability [38]. Given the sensitivity of logistic regression to the scale of input features, all numeric predictors in the training and test sets were standardised using z-score normalisation (centring and scaling using the `caret` package) [39]. The standardised training data was combined with the binary response variable 'IsPeak' and the model was fitted using the `glm()`function, incorporating all available predictors.

Model predictions on the test set were obtained as probabilities, at a threshold of 0.5 to yield binary classifications. Performance was assessed using a confusion matrix and further evaluated with a receiver operating characteristic (ROC) curve and the area under the curve (AUC) metric.

RANDOM FOREST & RECURSIVE FEATURE ELIMINATION (RFE)

To reduce dimensionality and improve model performance, RFE was employed using an RF classifier. RFE iteratively removes the least important features based on model performance until the optimal subset is identified [40]. A 5-fold cross-validation strategy was used to evaluate performance across varying subset sizes (1 to 40 features), using the `rfeControl` function, thus reducing overfitting [41]. The optimal feature set was then used to retrain a refined RF model, which was configured with 100 trees (`ntree = 100`), and the number of features considered at each split was set to the square root of the number of predictors (`mtry = 5`) [29]. Feature importance was computed and visualised using the Gini impurity index. The RFE-optimised model was evaluated on a test set and predictions were assessed using a confusion matrix. Feature importance scores were extracted from the trained RF model and ranked based on the Mean Decrease in Gini.

To evaluate the added value of feature selection, a baseline RF model was trained using the full set of features, with the same hyperparameters. Both the baseline and RFE-optimised models were compared in terms of accuracy, AUC, sensitivity and specificity, offering a multidimensional view of performance. To further validate model robustness, out-of-Bag (OOB) error rates were plotted against the number of trees. Moreover, to validate class separability, a density plot of predicted class probabilities was created to visualise the distinction between "Peak" and "No Peak".

To further optimise performance, hyperparameter tuning was conducted using the `caret` package and the `ranger` implementation of RF. A grid search was applied over combinations of `mtry ∈ {3, 5, 7, 9}` and `min.node.size ∈ {1, 3, 5}`, using 5-fold cross-validation to maximise the AUC-ROC score. Class probabilities were enabled to support probability-based performance metrics, and variable importance was assessed via permutation methods. The best-performing model from this tuning process was retained as the final model.

COMPARING MODEL PERFORMANCE

To evaluate the relative performance of the previously developed logistic regression model against the RF classifier, a model comparison summary table was compiled.

PREDICTING PEAK PERFORMANCES IN CURRENT PLAYERS

To forecast potential peak performance among current players, the final RFE-optimised model was used. The current player dataset was pre-processed to align with the model’s feature structure. This involved removing identifier columns like ‘Player’ and ‘Tournament.Year’ and handling any missing values by imputing them with zeros.

The processed data was passed through the trained RFE-optimised model to generate class probability estimates. Specifically, the predicted probability of a player achieving a peak performance in an upcoming season was extracted for each record. To focus on the most relevant predictions, the dataset was filtered to retain only the entries from the most recent tournament year. From this subset, the top 10 players with the highest predicted probabilities of peak performance were identified as "players to watch." These individuals were highlighted along with their corresponding predicted probabilities and key metrics to provide contextual performance metrics.

Through rigorous preprocessing, careful model selection, and comprehensive validation strategies, this analytical framework provides a robust basis for forecasting peak performances in professional T20 cricket. This approach extends traditional sport performance modelling by integrating domain-specific peak definitions, temporal lag features, and a tailored feature selection pipeline, reflecting both innovation and practical relevance.

RESULTS

DESCRIPTIVE STATISTICS

The clean dataset comprised of 223 unique IPL players, with a total of 501 player-seasons recorded between the years 2016 and 2019. Table I presents summary statistics for key batting performance metrics, providing insight into the dispersion and central tendencies of features.

TABLE I.

SUMMARY STATISTICS FOR KEY BATTING METRICS (2016-2019)

Metric	Average	Standard Deviation	Median	Skewness	Kurtosis	Q1	Q3
Runs Scored	145.88	168.52	73.00	1.41	1.80	14.00	243.00
Batting Average	20.23	15.33	18.33	0.92	0.98	7.16	29.38
Strike Rate	113.93	45.46	122.22	-0.59	0.50	90.00	141.88
4s (Fours)	13.00	16.48	5.00	1.54	1.94	1.00	20.00
6s (Sixes)	5.96	7.90	3.00	1.96	4.51	0.00	9.00
Fifties	0.82	1.46	0.00	2.25	5.67	0.00	1.00

Additionally, histograms of key variables (Figure 1) reveal that runs scored, number of 4s, and number of 50s are right-skewed, reflecting a large proportion of low-performing player-seasons, with a few players scoring substantially higher. The strike rate histogram shows a more symmetric distribution.

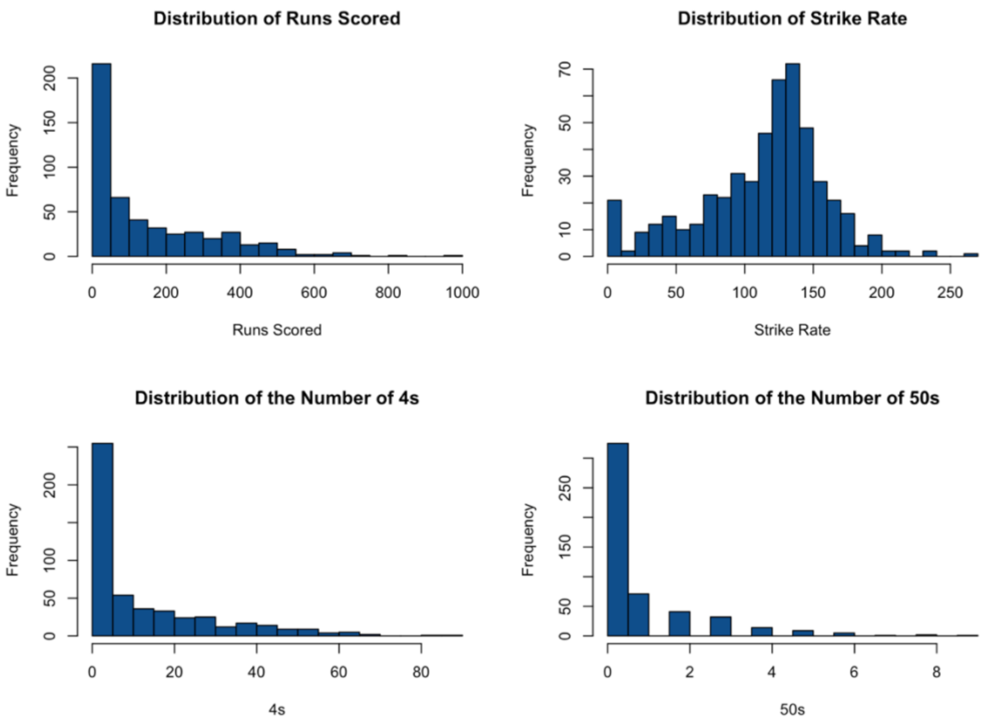

Fig. 1. Histograms of feature distributions

A correlation matrix (Figure 2) was generated to assess linear relationships among numeric variables. Runs scored was strongly correlated with balls faced, number of 50s, number of 4s, and number of 6s, as expected. Batting average and strike rate showed a moderate positive correlation, indicating that high-performing batters also tend to be more aggressive scorers.

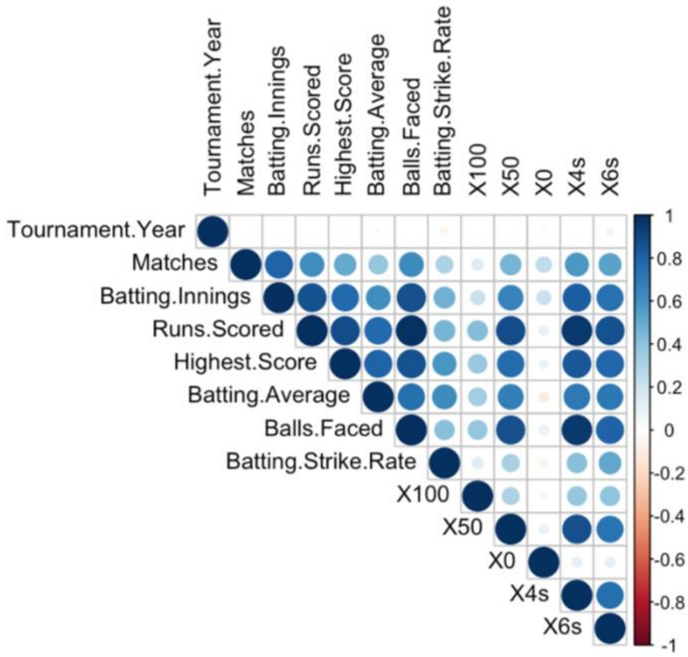

Fig. 2. Correlation Matrix of features

COMPARISONS WITH PEAK PERFORMANCE

Following computation of “Peak” seasons, an initial comparison was conducted between all player-seasons and their corresponding peak seasons. The average batting strike rate across all player-seasons was 113.93, while the average strike rate during peak seasons was 122.28, indicating a 7.33% improvement during peak years.

When comparing “Peak” versus “Non-Peak” seasons, there was an increase in strike rate during peak seasons, with a higher median value and greater variability. Batting average shows greater variability in peak seasons; however, the median appears to be lower. Similarly, the number of 4s shows a decrease in peak seasons (Figure 3).

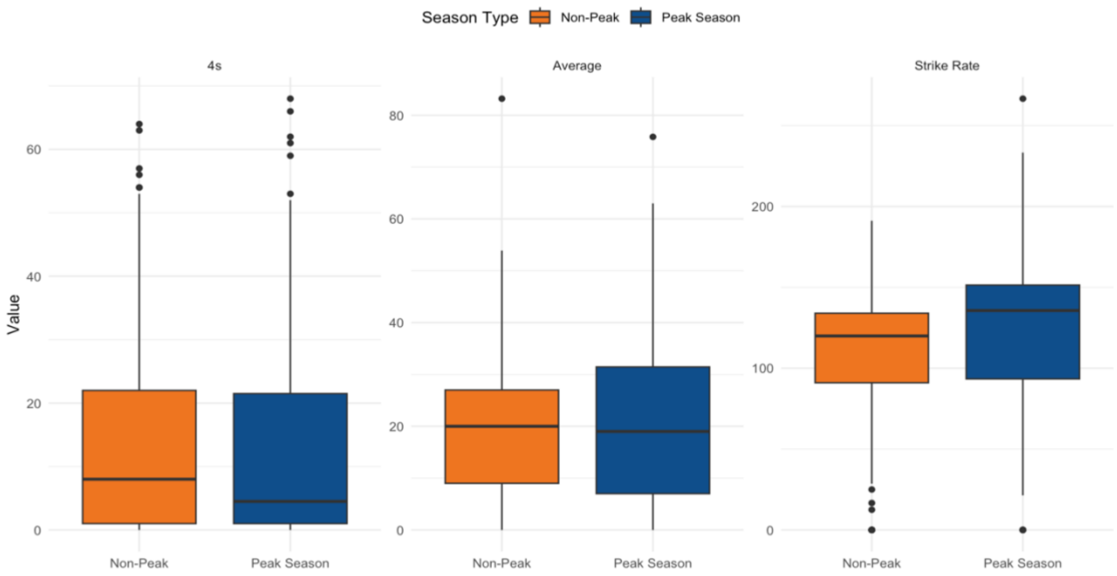

Fig. 3. Boxplots to illustrate differences between “Peak” and “Non-Peak” seasons

LOGISTIC REGRESSION

A logistic regression model was developed using engineered features from players’ historical performance data to act as a baseline. Model performance on the test set achieved an accuracy of 71.43%, with a 95% CI [0.60, 0.81]. The model’s sensitivity was 72.09% and specificity was 70.59%; the AUC was 0.76. A summary of the results can be seen in Table II. Balls faced and 50s scored had a negative association with peak probability ($p<.01$). Batting average trend, and trend in number of 50s were positively significant ($p<.05$). Previous season’s strike rate was negatively significant ($p<.05$).

TABLE II.

LOGISTIC REGRESSION SUMMARY

	Estimate	**Standard Error**	**Z value**	**Pr(>|z|)**
(Intercept)	-0.57	5.08	-0.11	0.91
Matches	-0.24	0.27	-0.91	0.36
Batting.Innings	-0.23	0.64	-0.36	0.72
Runs.Scored	4.43	4.85	0.91	0.36
Highest.Score	-0.72	0.50	-1.44	0.15
Batting.Average	-0.05	0.48	-0.11	0.91
Balls.Faced	-6.58	2.30	-2.87	0.004 **
Batting.Strike.Rate	0.48	0.33	1.47	0.14
X100	0.36	0.46	0.79	0.43
X50	-2.41	0.89	-2.72	0.006 **

X0	0.12	0.22	0.54	0.59
X4s	3.40	2.43	1.40	0.16
X6s	0.78	1.40	0.56	0.58
Peak.Year	-0.22	0.16	-1.33	0.18
prev1_Runs.Scored	0.06	4.00	0.01	0.99
prev2_Runs.Scored	3.67	2.21	1.66	0.10
trend_Runs.Scored	1.20	2.43	0.49	0.62
prev1_Batting.Average	1.63	0.85	1.93	0.05
prev2_Batting.Average	0.02	0.87	0.02	0.98
trend_Batting.Average	1.10	0.51	2.15	0.03 *
prev1_Batting.Strike.Rate	-1.49	0.70	-2.132	0.03 *
prev2_Batting.Strike.Rate	-0.82	0.73	-1.114	0.27
trend_Batting.Strike.Rate	0.30	0.39	0.78	0.43
prev1_X100	-0.21	0.57	-0.37	0.71
prev2_X100	1.37	63.12	0.02	0.98
trend_X100	-0.33	0.66	-0.49	0.62
prev1_X50	1.14	1.15	0.99	0.32
prev2_X50	0.43	0.69	0.62	0.53
trend_X50	1.82	0.82	2.22	0.03 *
prev1_X0	0.25	0.37	0.70	0.49
prev2_X0	0.30	0.25	1.19	0.23
trend_X0	-0.13	0.33	-0.38	0.70
prev1_X4s	-0.44	2.76	-0.16	0.87
prev2_X4s	-3.21	1.71	-1.87	0.06
trend_X4s	-1.80	1.80	1.00	0.32
prev1_X6s	-0.78	1.44	-0.54	0.59
prev2_X6s	-0.91	0.74	-1.23	0.22
trend_X6s	-0.61	1.14	-0.54	0.59
career_runs	0.06	1.44	0.04	0.97
career_innings	-0.39	1.27	-0.31	0.76
career_avg	0.21	0.74	0.29	0.77
experience	0.32	0.57	0.55	0.58

*(Notes: **p<.01 *p<.05)*

RANDOM FOREST & RFE

RFE was performed using an RF estimator and the highest model accuracy of 78.08%, with a moderate agreement of κ=.56 was achieved with a subset of 29 features (Figure 4). The top five most important predictors identified were: 'trend_Batting.Strike.Rate', 'Batting.Strike.Rate', 'trend_Batting.Average', 'Matches', and 'prev1_Batting.Strike.Rate'.

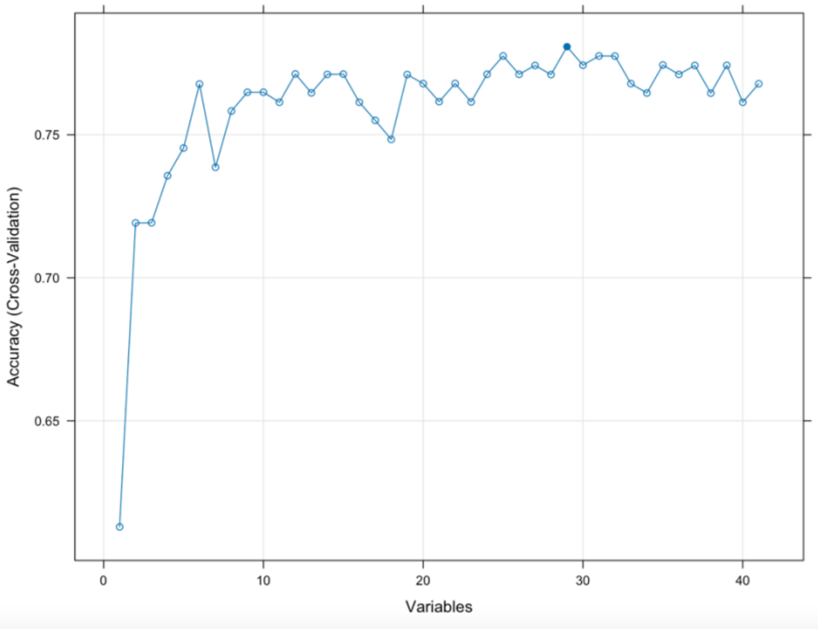

Fig. 4. Plot to illustrate optimal number of features

An RF model was retrained using the 29 RFE-selected features and revealed an accuracy of 75.32% with 95% CIs [0.64, 0.84]. The model had precision of 80.00% and an F1 score of 77.11%. Model sensitivity was 74.42% and specificity was 76.47%. As shown in Figure 5, the model correctly classified 26 positive cases and 32 negative cases, while misclassifying 19 instances in total.

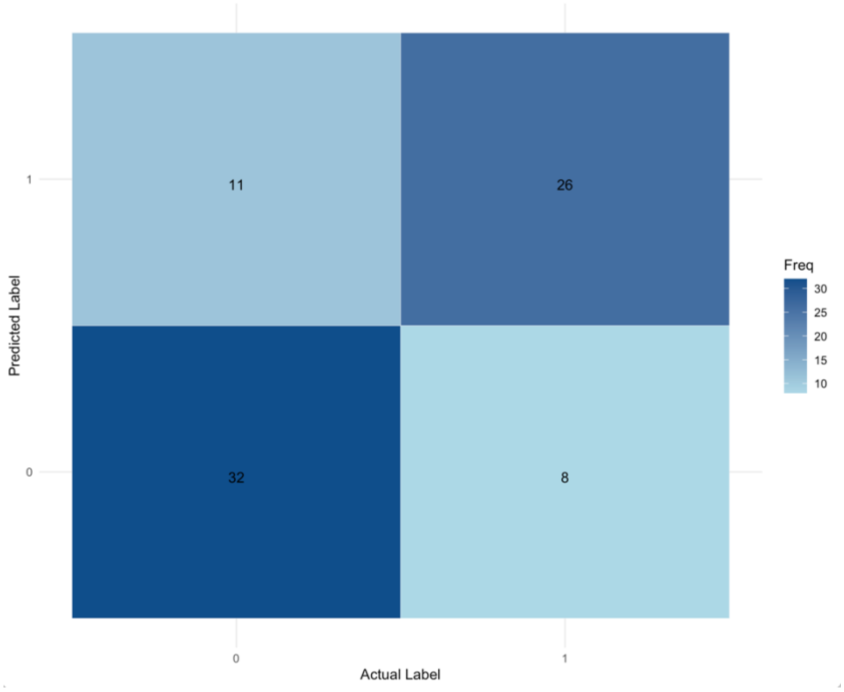

Fig. 5. Confusion Matrix of RFE-optimised model

The RFE-optimised model achieved an AUC of 0.84. Feature importance based on Mean Decrease in Gini index revealed that the most influential variables in the RFE-optimised model were 'trend_Batting.Strike.Rate', 'Batting.Strike.Rate', 'Highest.Score', 'Batting.Average', and 'Matches' (Figure 6).

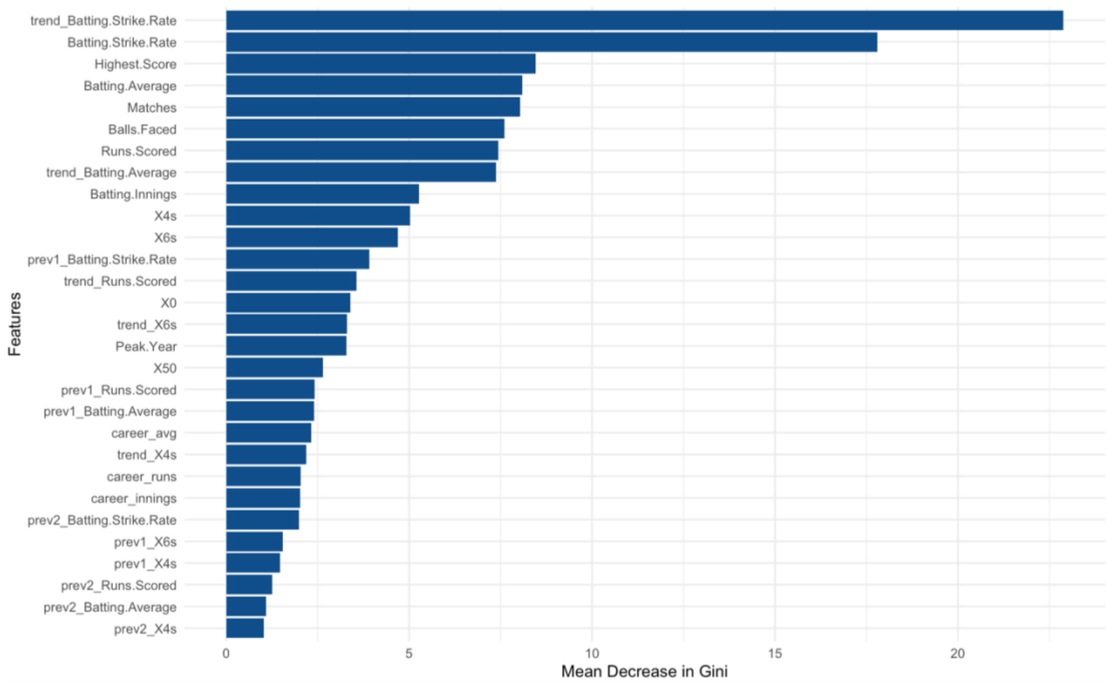

Fig. 6. Bar chart illustrating feature importance

For comparison, a baseline RF model was trained using all available features (prior to RFE). Compared to the baseline, the RFE-optimised model demonstrated improved accuracy

(75.32% compared to 72.72%), sensitivity (74.41% compared to 69.77%), and a slightly higher AUC (0.84 compared to 0.83), while specificity stayed the same at 76.47%.

The plot of the OOB error rate against the number of trees (Figure 7), showed a decline in error rate as the number of trees increased.

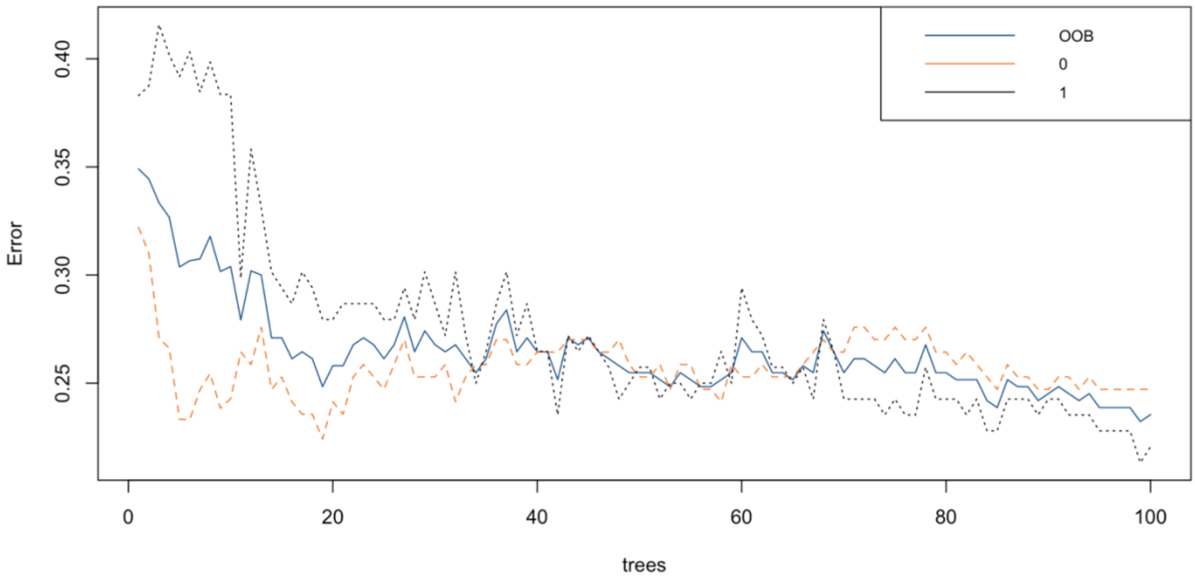

Fig. 7. Out-of-bag (OOB) error rate versus number of trees

The results of a Welch Two Sample t-test revealed a statistically significant difference between the predicted probabilities for the two classes, "Peak" and "Non-Peak" $t(74.55)= -6.49$, $p<.001$. The mean predicted probability for the "Peak" class (0.62) was significantly higher than that for the "Non-Peak" class (0.30), with a 95% CI [-0.43, -0.23].

After conducting a 5-fold cross-validation grid search to tune the hyperparameters, the best performing model was identified with the parameters mtry = 5, min.node.size = 3, and splitrule = gini. This configuration yielded the highest AUC of 0.84, sensitivity of 77.06% and specificity of 78.70%.

COMPARING MODEL PERFORMANCE

Both the logistic regression and tuned RF model exhibited similar sensitivity; however, the RF model outperforms the logistic regression model with higher accuracy, AUC, specificity and precision (Table III). A visual comparison of ROC curves (Figure 8) shows the RF curve as consistently above the logistic regression curve across all thresholds.

TABLE III

LOGISTIC REGRESSION AND RANDOM FOREST MODEL COMPARISON

Model	Accuracy	AUC	Sensitivity	Specificity	Precision
Random Forest	0.75	0.84	0.72	0.79	0.82
Logistic Regression	0.71	0.76	0.72	0.71	0.76

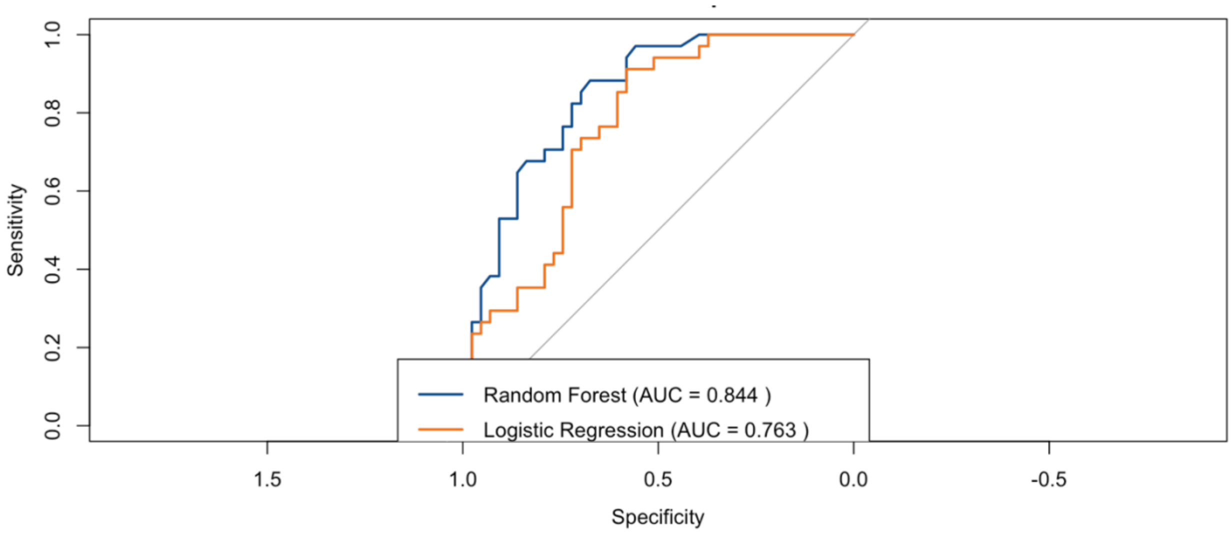

Fig. 8. ROC curves for random forest and logistic regression models

PREDICTING PEAK PERFORMANCES

The top 10 players with the highest predicted probability of peaking were identified (Table IV).

TABLE IV

TOP 10 PLAYERS PREDICTED TO PEAK

Player	Peak Probability	Runs Scored	Batting Average	Strike Rate
SD Lad	0.97	15	15.00	115.38
P Simran Singh	0.96	16	16.00	94.11
N Rana	0.94	344	34.40	146.38
Gurkeerat Singh	0.93	98	32.66	140.00
MJ Guptill	0.93	81	27.00	152.83
S Gopal	0.92	63	15.75	136.95
TA Boult	0.92	7	7.00	175.00
CH Gayle	0.91	490	40.83	153.60
KA Pollard	0.91	279	34.87	156.74
MJ Santner	0.91	32	32.00	139.13

CONCLUSION

SUMMARY & IMPLICATIONS

This study aimed to predict whether a given season represents a batter's peak performance in the IPL using ML models. The analysis focused on strike rate as the primary indicator of peak performance, given its critical role in the fast-paced T20 format [9], [11].

Findings revealed that during peak seasons, batters exhibited a 7.33% higher strike rate compared to non-peak seasons. Fluctuations in other indicators, like batting average and boundary count, highlighted the multidimensional nature of player performance. This emphasises that performance should not be judged based on a single metric. It also means that coaches should tailor training based on players' individual needs.

Among the models evaluated, the RF algorithm, optimised through RFE, achieved the highest predictive accuracy at 75.32%. However, this accuracy is lower than that reported in previous cricket studies, which may be attributed to their use of weighted models for assessing variable importance [24]. Moreover, the most informative features included recent strike rate trends, historical strike rate, and batting average; these metrics should be monitored closely to gain the best insights into future performance.

The RF model successfully identified players likely to enter peak form, providing actionable insights for recruitment and team selection strategies. Players demonstrating upwards trends in strike rate and consistently high scoring were flagged as strong peak candidates. For example, N. Rana may appeal to coaches for recruitment, as the model predicts he will peak with both a high strike rate and run rate. These results validate the hypothesis that ML can detect meaningful patterns in performance trajectories, reinforcing its value in talent identification. This aligns with previous research emphasising the increasing role of data-driven approaches in professional sport [3], [7-9], [12], [15], [23], [24], [26], [28], [32], [33].

Coaches could leverage similar models to monitor player development and align peak performance windows with crucial phases of the tournament, such as utilising aggressive batters in "death overs" [42]. These results may also help stop organisations from overinvesting in players who are past their prime, while uncovering late bloomers ahead of their competitors. This model can adapt across formats (e.g., ODIs, tests) by adjusting key performance metrics.

LIMITATIONS

This study's primary limitation lies in its dataset, which only included 2016-2019 IPL seasons and excluded broader performance dimensions such as bowling and fielding. The sample may also introduce bias where some players played fewer seasons. Consequently, the model may not fully capture a player's overall value, especially in T20 formats, where fielding has become increasingly critical and the ability to save runs is crucial for winning games [43]. However, it is difficult to quantify fielding ability [32]. Notably, external factors such as injuries [44], weather [45], and team dynamics [46] play a critical role, so incorporating these data could further enhance predictive accuracy.

Furthermore, defining peak performance based solely on strike rate, may oversimplify complex player dynamics, potentially overlooking key contextual elements like consistency or match situations. Incorporating these factors in future studies could significantly improve predictive accuracy.

FUTURE DIRECTIONS

Future research should aim to enhance the depth of analysis by expanding the dataset to include additional seasons, leagues, and a broader set of performance metrics, such as physiological data, and contextual match conditions. This would improve the generalisability of findings and allow for a more nuanced understanding of player performance [47]. However, ethical issues may arise with the collection of biometric athlete data, and these must not be ignored [48]. A more inclusive indicator of peak performance should also be developed that integrates traditional metrics like strike rate and batting average, with situational impact through weighted scoring models, to provide a more accurate measure of peak performance.

If larger datasets become available, the application of advanced ML techniques, such as neural networks, could be explored to assess comparative player performance more effectively [32], [33]. Further research could also focus on constructing a dynamic player ranking framework, leveraging algorithms like SVMs because of their powerful classification abilities, especially in situations that need fine-grained attribute weighting rankings [3].

Finally, developing simulation tools to predict player performance under varying match conditions, as demonstrated in football [49], could significantly enhance decision-making processes. Integrating predictive models into coaching workflows can improve player selection, training focus, and tactical decisions, effectively translating data insights into actionable strategies to optimise performance outcomes.

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